

Comprehensive Electricity Demand Forecasting With a Custom Multi-Dimensional Dataset With Model Analysis and Mobile Visualization

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ABSTRACT Accurate electricity demand forecasting is critical for efficient energy management in regions affected by rapid urbanization, grid instability, and load shedding. The traditional forecasting methods are inadequate when it comes to the short-term variation due to weather conditions, socio-economic activities, and unforeseen events. This paper introduces a *deployable, statistically grounded electricity demand forecasting model* that incorporates meteorological, calendar-based, and grid operational information, such as temperature, humidity, rainfall, daylight hours, festive days, past consumption, and load shedding data. To learn about the complex non-linear and temporal relationships in daily electricity demand, Random Forest and Long Short-Term Memory (LSTM) frameworks are used and to facilitate the implementation of the model in *near real-time*, the framework uses a FastAPI-based backend and a Flutter-based mobile app to visualize the predictions. *Resistance to sensor noise and non-observable types of events* is improved by integrating contextual features and outlier-aware data preprocessing. The experimental results on a dataset with a time span of 7 years (2017–2024) indicate that the LSTM model is superior to the traditional models, where the RMSE is reduced from 7.43 to 4.84 and the MAPE from 14.76% percent to 8.2%. The proposed system provides a useful and scalable solution in the dynamic power grid environment for real-time electricity demand forecasting.

INDEX TERMS Energy, urbanization, load-shedding, LSTM, mobile application.

I. INTRODUCTION

Load shedding affects the economic operations, which vary in magnitude in different parts of the world. In India, the fuel shortage resulted in up to 12-hour blackouts in the 2022 energy crisis [1], [2]. The power outages are 6 hours a day in

South Africa due to a 14-year power crisis, whereas the power outages in Pakistan are 12-15 hours in average, aggravated by the reliance on imported fuel sources [3], [4]. There are also long power outages in Sri Lanka and Lebanon caused by the breakdown of the energy systems, and Bangladesh is

TABLE 1. Load Shedding Statistics in Different Countries [1], [3]

Country	Daily Duration	Primary Causes	Impact
India	10-12 hours	Fuel shortages, high demand	Power plants shut down, major blackouts in industrial and residential areas
South Africa	Up to 6 hours	Insufficient generation capacity, high demand	Businesses face disruptions, prolonged economic slowdown
Pakistan	12-15 hours	Imported fuel dependency, rising costs	Urban and rural areas suffer, economic downturn
Sri Lanka	Daily outages	Hydropower failure, fuel shortages	Economic crisis worsens, protests erupt
Lebanon	22-23 hours	Economic collapse, fuel shortages	State electricity available for only 1-2 hours daily, reliance on private generators
Bangladesh	Varies (up to 1500 MW shortage)	Energy import costs, overcapacity issue, gas depletion	Urban-rural divide worsens, CMSMEs affected, agricultural losses

struggling with the power shortage of 1500 MW, which is a heavy burden on the rural land and agricultural integration [1], [5]. A summary of these challenges is presented in Table 1.

Effective energy management is essential for economic development. With global power demand expected to grow at 2.1% annually from 2025 to 2040, driven by urbanization, this rise will inevitably lead to challenges such as grid instability and economic losses due to wasted energy [6]. Conventional statistical models like ARIMA and linear regression are often inadequate in capturing the nonlinear and dynamic nature of electricity demand [2], [4], [7]. While machine learning methods offer more flexibility and precision [5], [8], [9], many existing models focus on long-term predictions and fail to account for real-time fluctuations in demand, particularly in the context of load shedding [10]. One major challenge remains the integration of Demand Response (DR) programs with real-time forecasting, as DR requires accurate short-term predictions to adjust generation, storage, and consumption patterns dynamically. However, traditional models struggle to capture socio-economic factors, weather variations, and unexpected grid events, which are essential for effective DR execution [11]. To address this gap, we propose a data-driven forecasting framework that supports DR operations, enhances grid reliability, and minimizes the risk of load-shedding. The recent research investigated further deep learning models regarding electricity load prediction such as attention-based and Transformer models capable of sufficiently addressing long-range temporal correlations in time-series data. Transformer-based forecasting models have shown promising results in terms of performance because they can process sequential data without using recurrent structures. Moreover, hybrid methods that use traditional machine learning models with deep neural networks have become popular due to their capability to combine nonlinear feature association with time learning mechanisms. These advancements underline the need for smart forecasting models that can address the challenges of dynamic electricity demand patterns in today’s electricity systems. Driven by these developments, this paper explores machine learning as well as deep learning methods of electricity demand prediction on a multi-dimensional dataset.

This paper presents a data-driven framework for daily electricity demand prediction using Random Forest (RF) and

Long Short-Term Memory (LSTM) networks. Our approach models intricate consumption behaviors and the key factors influencing electricity demand prediction [12]. The system also integrates a mobile application for real-time monitoring and adaptive responses to demand fluctuations [13].

The key objectives and findings of this study are as follows:

- *Objective 1:* To develop an efficient predictive model that incorporates weather, socio-economic factors, and historical consumption patterns.
- *Objective 2:* To apply RF and LSTM models to capture the complex, non-linear characteristics of electricity demand.
- *Objective 3:* To evaluate the model’s performance using MAE, MAPE, RMSE, and R-squared, showing significant improvements over traditional methods.
- *Objective 4:* To design a dynamic and flexible framework, with a mobile app to visualize predictions in real-time.

This paper provides an efficient and scalable solution for managing daily electricity consumption in rapidly developing regions, with a focus on improving real-time forecasting capabilities and supporting Demand Response programs.

The remaining part of the paper is provided as follows: **Section 2** is the background and fundamentals; **Section 3** is reviews and related works; **Section 4** is describing the methodology; **Section 5** is discussing experimental results and comparisons and **Section 6** is conclusion in future direction.

II. LITERATURE REVIEW

The necessity of electricity demand forecasting is becoming very critical because of urbanization and energy reliance and proper forecasting will ensure stability in the grid and long term energy management. As opposed to conventional statistical models which have been in wide use, ML and DL are more appropriate in modeling nonlinear and dynamic consumption patterns. Previous research such as [2], [4], [5], [14], primarily used statistical and end-use based methods that do not necessarily appreciate short term fluctuations as a result of weather, socio-economic and load shedding. Lee and Cho [15] compared conventional and ML-based models, showing that LSTM outperforms traditional methods in peak load forecasting. Similarly, Velasquez et al. [16]

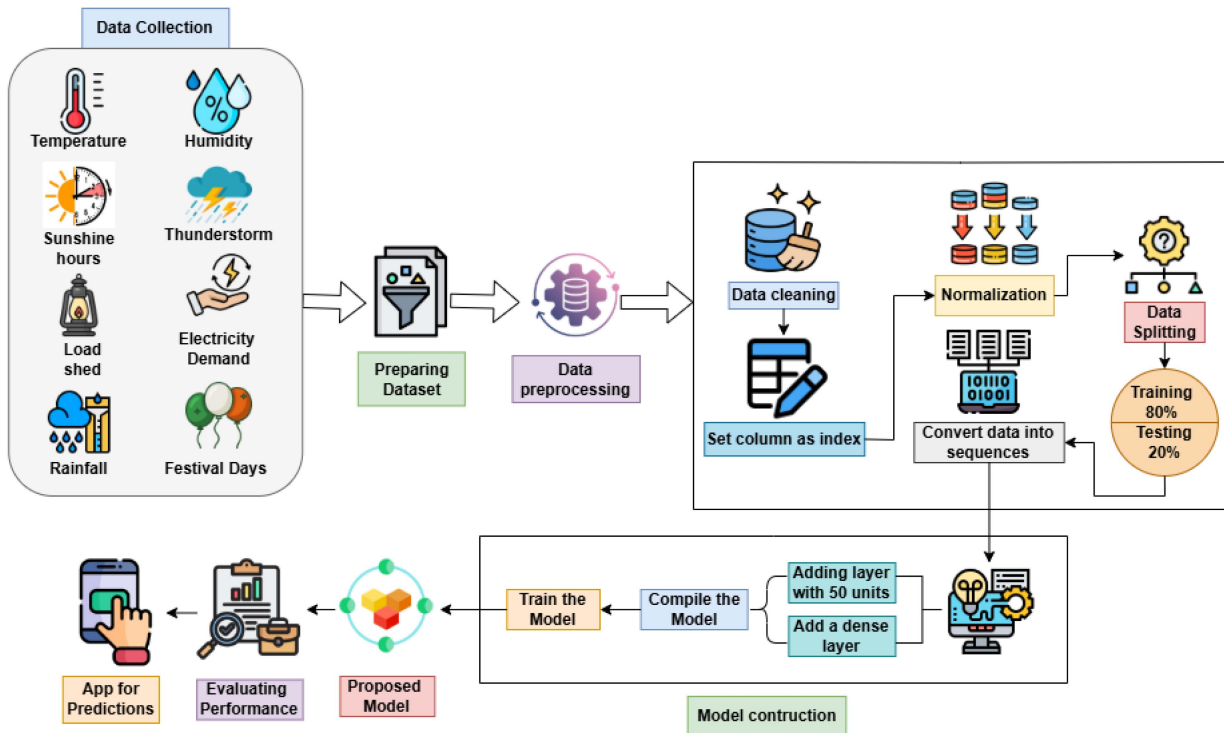


FIGURE 1. Proposed methodology for the electricity demand prediction system. The diagram illustrates the overall workflow from data collection to real-time prediction.

highlighted the effectiveness of hybrid time series models that account for seasonality. Tang et al. [8] integrated ML and blockchain for predictive energy trading in microgrids, enabling real-time forecasting. Alquthami et al. [17] found that decision tree models, particularly those with gradient boosting, outperformed other classifiers in smart grid forecasting.

In isolated power systems, Matrenin et al. [18] demonstrated that ensemble models combining SVR and random forests reduce overfitting and improve accuracy. LSTM networks, as used by Torres et al. [19], achieved under 1.5% error in short-term forecasting with hyperparameter optimization.

Recent work explores CNNs for feature extraction: Yazici et al. [20] showed that 1D CNNs can rival recurrent models, while Khan et al. [21] proposed a CNN-LSTM framework that outperformed baseline models. Comparative studies by Bilgili and Pinar [22] confirmed LSTM's superiority over SARIMA in Türkiye, whereas Nazir et al. [23] introduced the Temporal Fusion Transformer (TFT), which improved accuracy but required complex tuning.

Hybrid and advanced models also show promise. Abumohsen et al. [24] found GRUs performed best among recurrent networks. Mounir et al. [25] used BI-LSTM with EMD for high-accuracy forecasting, albeit with high computational cost. Fan et al. [26] combined EWT, CNN, RNN/LSTM, and BOA, and González Grandón et al. [27] integrated ARIMA with LSTM and regression, both achieving high accuracy but with increased model complexity. El-Azab et al. [28] created

a seasonal forecasting system based on ANFIS, LSTM and GRU and ANN that has been shown to be effective in integrating the relationship between temperature and forecasting.

However, there are numerous studies that do not consider such important factors as load shedding, grid reliability, and socio-economic conditions limiting real-world scenarios. To counter this, we suggest a hybrid RF-LSTM scheme to capture non-linearities and temporal patterns with real-time grid and load-shedding data, and the outcomes are visualised using a mobile application to implement the proposed system in rapid urbanizing areas.

III. PROPOSED SYSTEM MODEL AND METHODOLOGY

The proposed system uses a process flow for electricity demand prediction, as shown in Fig. 1. The process starts with the collection of data from various sources, such as historical electricity demand, weather factors (e.g., temperature, humidity), and socio-economic factors (e.g., holiday calendar, population density). The data is then preprocessed and feature engineered by cleaning the data (e.g., removing missing values), normalising the data, and selecting features. The preprocessed data is then processed through machine learning algorithms, namely, Random Forest (RF) and Long Short-Term Memory (LSTM) networks, which can capture linear and non-linear patterns in electricity consumption. The system is expected to adapt to different patterns by using these models to forecast electricity consumption based on historical data and other factors such as weather and socio-economic

TABLE 2. Overview of Experimental Dataset Attributes

Attributes	Description
Number of Columns	11
Number of Rows	3105
Independent Features	Temperature, Humidity, Daytime duration, Rainfall, Electricity demand, Load shed
Categorical Features	Thunderstorm, Hazardous weather, Festival day
Target Feature	Electricity Demand

events. Additionally, the system is equipped with a mobile app interface for monitoring demand variations. This application helps utilities to dynamically adjust to demand variations through real-time predictions and alerts. The model is evaluated using performance metrics: MAE (Mean Absolute Error), RMSE (Root Mean Square Error), MAPE (Mean Absolute Percentage Error) and R2. These criteria guarantee the demand forecasting predictions are accurate. Fig. 1 provides a glimpse into the proposed approach, showcasing the implementation of forecasting and demand response using the data in the proposed system. This depicts the entire workflow from data collection to data processing and prediction and real-time adaption of electricity demand.

A. DATASET DESCRIPTION

1) DATA SOURCE

This dataset includes the data on historical electricity demand, weather, and contextual data of the three sources namely:

- *Meteorological Data*: Collected from BMD portal [29].
- *BPDB Loads*: Daily electricity demand, peak loads and load-shedding records of Bangladesh Power Development Board [30].
- *Time and Date Website*: Calendar holidays and festival data of Time and Date [31].

The total contribution of all the sources is given in Table-2 illustrates the daily values of temperature, humidity, thunderstorms, daytime duration, rainfall, festive days, historical electrical demand, and load shedding in the period between January 1, 2017 and June 31, 2024. To ensure experimental reproducibility, the assembled data combines publicly available data on BMD, BPDB, and TimeandDate portals. The data range covers the interval between January 1, 2017 and June 30, 2024, which has adequate seasonal variations to train and test. To preserve the temporal integrity of forecasting, the dataset was divided chronologically into training (2017–2021), validation (2022) and testing (2023–mid 2024) subsets. The time-dependent splitting method aids in avoiding data leakage and enables the models to generalize on seasonal demand changes and possible concept drift due to shifting consumption patterns over time.

2) DATA FEATURES

The following data set is used in the study to understand the factors affecting daily electricity demand, both meteorological and non-meteorological factors:

- *Date (DD-MM-YY)*: Shows the day of the month on which the data was being collected for each year.
- *Temperature (°C)*: Integral fixed value relating to the average temperature on the day analyzed, expressed in Celsius degrees.
- *Humidity (%)*: Persistent numeric variable, which measures the air relative humidity in terms of the percent.
- *Sunshine (Hours)*: Discrete variable, which measures the quantity of Elementary sunlight, that the patient was exposed to, in hours for the day.
- *Night-time (Hours)*: Discrete variable, or total amount of time of darkness throughout the day.
- *Rainfall (mm)*: Continuous variable, representing a total amount of the precipitate of the day in millimeters.
- *Hazardous Weather (1/0)*: Categorical variable, it measures whether a location experiences risky weather conditions, including torrential rainfall or powerful wind, in one day or not.
- *Thunderstorm (1/0)*: Measuring the probability of thunderstorms, and since it is a categorical variable with two categories – the presence of thunderstorms and no thunderstorms – we assign it the value of 1 and 0, respectively.
- *Festival Day (1/0)*: Categorical variable, representing whether the day is a festival or not; where 1 signifies yes for festival and 0 for no.
- *Demand (MW)*: Steady variable referring to the overall energy consumption for the certain day and described in megawatts (MW).
- *Load Shed (MW)*: Quantitative data with the number of load shedding (unserved demand) for the day in megawatts (MW).

Due to this diverse range of variables, it becomes possible carrying out an evaluation on the determinants of electricity demand inclusive of weather conditions, societal activities and the performance of the grid. Categorical variables like, *Thunderstorm*, *Hazardous Weather*, and *Festival Day* were encoded under binary encoding where 1 denoted the occurrence of an event and 0 denoted the non-occurrence of an event. This encoding method enables the machine learning models to effectively encode contextual variables like extreme weather conditions and social activities that can greatly impact the electricity demand patterns.

B. DATA PREPROCESSING

Constructive data preprocessing guarantees the reliability of the models through the improvement of the quality of the data, the processing of the inconsistencies, and the preparation of the time series forecasting data.

1) HANDLING MISSING DATA

The missing values in the data set are due to the logging error or the missing record of the sources such as BPDB and meteorological agencies, being addressed by:

TABLE 3. Comparison of the Proposed LSTM-Based System With Existing Forecasting Approaches

Study	Model Type	Input Features	Forecast Horizon	Evaluation Metrics	Key Observations
Proposed LSTM System	LSTM + RF	Historical demand, temperature, humidity, rainfall, calendar events	Daily	MAPE, RMSE, MAE, R ²	Captures non-linear and temporal dependencies, highest accuracy (MAPE 8.21%), adaptable to real-time updates
Random Forest [2]	RF	Historical demand, temperature	Daily	MAPE, RMSE	Handles non-linear relationships but limited in capturing temporal dependencies
ARIMA [4]	ARIMA	Historical demand	Daily	RMSE, MAE	Performs poorly for non-stationary and non-linear patterns
Prophet [5]	Additive Time Series	Historical demand, calendar events	Daily	MAPE, RMSE	Captures trends and seasonality but limited for complex non-linear dynamics
Gradient Boosting	GB	Historical demand, temperature, humidity	Daily	MAPE, RMSE	Effective for non-linear patterns, but less accurate for long-term dependencies

- **Forward Fill:** For temporal variables like electricity demand or temperature, assuming continuity in trends over short intervals.
- **Mean Imputation:** Applied to non-temporal features to preserve overall data distribution while minimizing information loss.

In addition to addressing missing value treatment, noisy observations due to sensor anomalies or irregularities in reporting were corrected by simple smoothing and validation checks. Temporal continuity of electricity demand data was maintained by filtering sudden spikes in the data due to recording anomalies prior to model training. These preprocessing steps enhance the reliability of the forecasting models when working with real-world monitoring data.

2) OUTLIER TREATMENT

Outliers due to power surges, extreme weather, or entry errors were detected using the Interquartile Range (IQR) method. Values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$ were normalized to reduce their influence during model training, ensuring robust learning without distortion from anomalies.

C. FEATURE SCALING (MIN-MAX NORMALIZATION)

Feature scaling is a critical preprocessing step in machine learning to ensure that all input variables contribute equally to the model's learning process. In this study, continuous features such as temperature, humidity, and electricity demand are scaled using Min-Max normalization, which transforms the features to a range of [0, 1]. The formula for this transformation is:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

Where x is the original feature value, and $x_{\min}$ and $x_{\max}$ represent the minimum and maximum values of the feature in the dataset. Scaling ensures that variables with large numerical ranges, such as electricity demand (in megawatts), do not dominate smaller-range features like temperature (in degrees Celsius). This equalizes the contribution of all features, enhancing model convergence during training. Feature scaling is particularly beneficial for models like Random Forest and Long Short-Term Memory (LSTM) networks, which are sensitive to the scale of input features. By stabilizing training

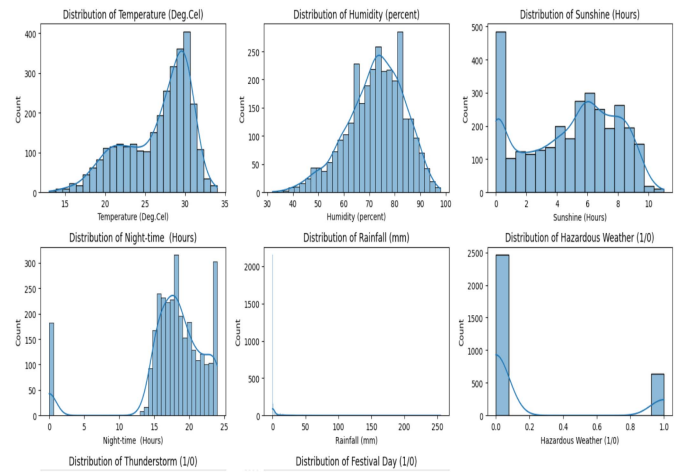


FIGURE 2. Histogram of numerical features for electricity forecasting. This image shows the distributions of the training features, including temperature, humidity, rainfall and daily electricity demand, which are important for the forecasting models. These features are essential to explain electricity demand and its seasonal and weather variability.

and improving prediction accuracy, feature scaling becomes an essential part of our preprocessing pipeline.

D. EXPLORATORY DATA ANALYSIS

In this section, the effect that the various features have on electricity demand will be analyzed, which will assist in determining significant predictors, and understanding of the model will be enhanced to improve the accuracy of predictions.

1) FEATURE DISTRIBUTION ANALYSIS

The distribution of features plays a crucial role in the performance of the models since the presence of patterns in the variables determines how a model will be able to learn, its accuracy to predict and how sensitive it is to outliers. The Fig. 2 shows important numerical characteristics, such as temperature, humidity, rainfall and daily electricity demand.

The temperature and humidity is also near normally distributed with seasonal extremes whereas rainfall is highly skewed with frequent low values and occasional extreme heavy precipitation which may lead to sudden demand peaks. The seasonal trend of electricity demand is vivid with high

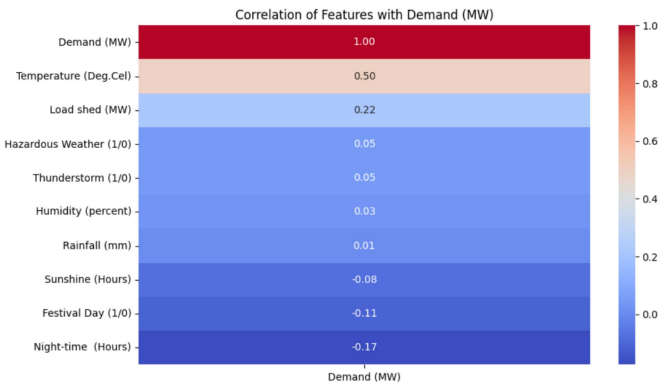


FIGURE 3. Correlation of features with electricity demand.

demands during the summer seasons because of cooling requirements and low demands during winter seasons because of weather and fluctuations in seasons.

The perceived seasonal patterns of demand imply the significant role of capturing the time-dependencies in forecasting models. The demand of electricity is higher in summer due to the high cooling demand and lower in winter due to the relatively low demand. The several years of historical data incorporated in the models enable them to learn repetitive seasonal trends and enhance their resilience to changes in electricity consumption over time.

These distributions are important to analyze in order to preprocess. The skewed variables are not allowed to affect the models disproportionately through the use of appropriate normalization, transformation, and outliers, which increase the strength and accuracy of both Random Forest and LSTM.

2) CORRELATION ANALYSIS

Pearson correlation was used to determine the strength of relations between features and their effect on demand. In Fig. 3, the heatmap shows that the extent of correlation between features and electricity demand.

Key findings include:

- *Temperature*: Moderate positive relation ($r = 0.50$), corresponding to the use of cooling systems more often on hot weather.
- *Humidity*: Weak positive correlation ($r = 0.03$), indicating not having any significant impact.
- *Rainfall*: Non-significant correlation($r = 0.01$), indicating weak effect.
- *Festive Days*: Weak negative correlation ($r = -0.11$), because of less industrial and commercial activity.

These correlations guide the choice of features as well as improve the performance of the forecast model.

3) FEATURE IMPORTANCE ANALYSIS

In order to gain more insight into the role played by each feature in the prediction, the feature importance analysis was conducted using the Random Forest model. Random Forest

estimates have an important property in which each feature is evaluated by the amount of impurity reduction achieved by multiple decision trees. This analysis aids in determining the most critical predictors that influence electricity demand. These findings demonstrate that meteorological variables, including temperature and seasonal indicators are predominant in demand forecasts, followed by contextual variables, including festival days and grid-related variables. The contributions of these features can also be used to enhance model interpretability and assist in decision-making by energy management systems.

E. MACHINE LEARNING ALGORITHMS

Electricity demand forecasting plays a vital role in the efficient operation of power systems, particularly in planning generation, reducing costs and enhancing energy security [32]. Machine learning approaches have shown their effectiveness in capturing the underlying nonlinear relationships in electricity demand data. Different methods have been used to predict electricity demand, each with its own merits depending on the dataset and temporal features of the electricity demand.

These models include traditional statistical and machine learning models such as Linear Regression and Polynomial Regression, to more sophisticated models such as Random Forest, Gradient-Boosting, Auto Regressive Integrated Moving Average (ARIMA) and deep learning models including Long short-term memory (LSTM) networks. The choice of model depends on the interpretability, accuracy and ability to capture time-series data characteristics of the model [33].

Random Forest and Long Short-Term Memory (LSTM) models are applied separately, not as a hybrid or ensemble model in this study. The objective is to evaluate their relative performance in terms of capturing various features of electricity demand data. Random Forest specifically works well with nonlinear relationships between multiple variables used as inputs, whereas LSTM networks are better suited to learning time-dependent and sequential relationships that exist in time-series data. Through the comparative analysis of these models, this research will offer a better insight into the relative advantages of the classical machine learning and deep learning methods to a multi-dimensional electricity demand forecasting dataset.

F. PROPOSED MODEL FRAMEWORK

This algorithm is most effective in time series forecasting scenarios and is highly encouraged in these cases. It is well capable of capturing the dynamic nature and the consequent relationships that exist between past trends and future trends. In this research, we adopt the Long Short-Term Memory (LSTM)-based method to forecast electricity demand, as shown in the Fig. 4.

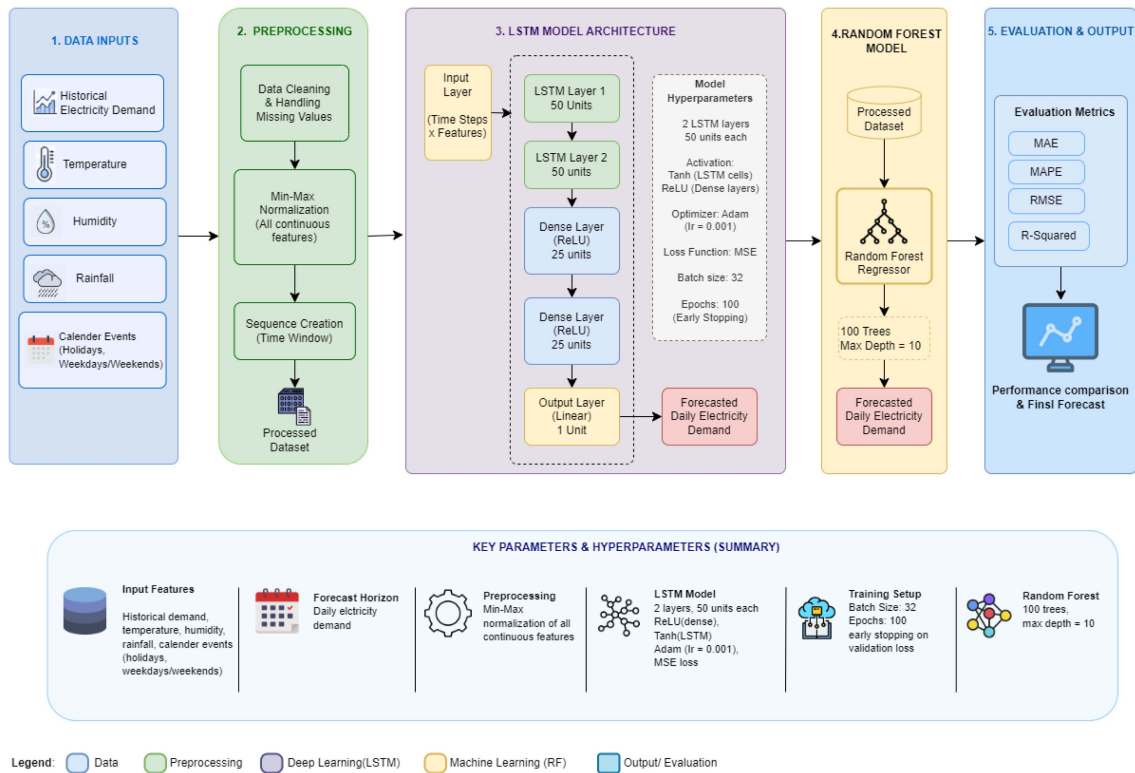


FIGURE 4. Proposed electricity demand forecasting system architecture. The framework combines various input characteristics, such as historical demand, weather conditions (temperature, humidity, rainfall), and calendar data, and then preprocesses it with data cleaning, Min-Max normalization and sequence generation. The processed data are subsequently inputted into a deep learning model of a two-layer LSTM network and a machine learning model of a Random Forest regressor. Lastly, the system measures performance of forecasting through MAE, MAPE, RMSE and R^2 to produce precise predictions of daily electricity demand.

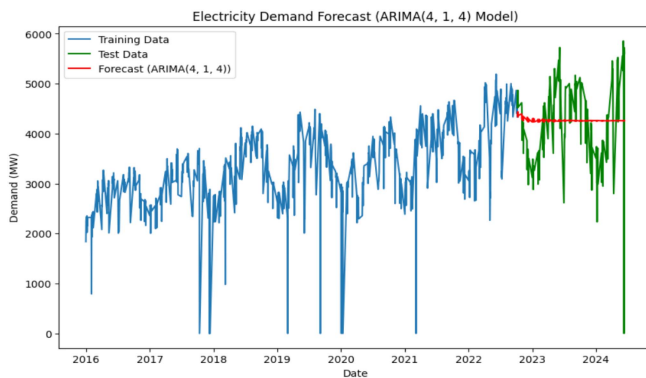


FIGURE 5. Forecast demand with ARIMA.

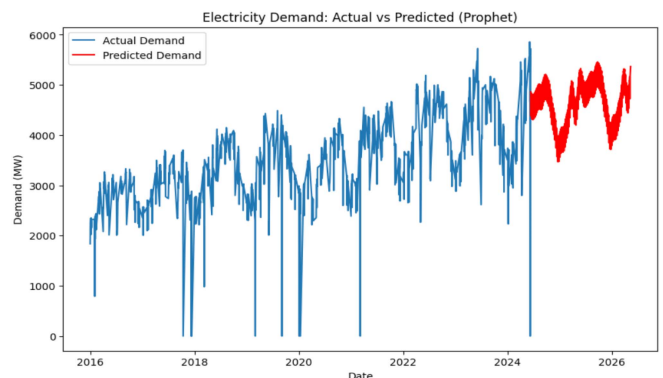


FIGURE 6. Forecast demand with Prophet.

1) INCORPORATION OF TIME DEPENDENCIES AND EXTERNAL VARIABLES

The LSTM model is trained using past electricity demand data along with meteorological and socio-economic factors such as temperature, humidity, rainfall, festive days, and load shedding records. These external characteristics are added to values of historical demand at each time step in order to give more contextual information and enhance the forecasting accuracy.

G. HYPERPARAMETER OPTIMIZATION

To achieve the best performance of the models, hyperparameter optimization was performed through a grid search approach with validation evaluation. In the case of the Random Forest model, the number of trees, the maximum tree depth and the minimum number of samples needed to split a node were systematically tested to identify the best configuration. In the same way, various hyperparameters of the LSTM model were experimentally optimized to enhance

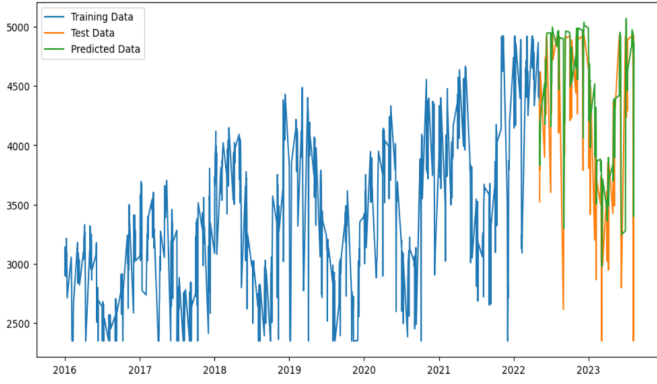


FIGURE 7. Forecast demand with LSTM.

forecasting. The tuning was done by experimenting with various values of learning rate, number of stacked LSTM layers, batch size and dropout rate. In particular, the learning rate was adjusted between 0.001 and 0.01, and the LSTM layers were also adjusted from one to three layers. The batch size of 16, 32 and 64 was tested to trade-off between training efficiency and model convergence. Moreover, dropout regularization was used between 0.1 and 0.5 to reduce overfitting in the training process. Adam optimizer was chosen because it has an adaptive learning rate and converges steadily. The final hyperparameter configuration of the two models were chosen according to the least validation error of the training process.

IV. EXPERIMENTAL RESEARCH AND ANALYSIS

A. EVALUATION METRICS

To evaluate the performance of the machine learning models for electricity demand prediction, several standard regression metrics were used.

- **R-Squared (R^2):** R-Squared is a measure of how much the predictor variables explain the variance of the dependent variable [34]. It is the percentage of variance for the observed data that is explained by the model, and is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the observed values.

- **Mean Absolute Error (MAE):** MAE is defined as the average absolute error (the absolute mean error) between the predicted value and true value [35]. It is defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value. Lower MAE indicates better accuracy.

- **Root Mean Square Error (RMSE):** RMSE is a measure of how close predictions are to the actual values. It is

calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

where y_i represents actual values and $\hat{y}_i$ represents predicted values. Lower RMSE indicates better performance.

- **Mean Absolute Percentage Error (MAPE):** MAPE is the percentage error of the predicted value, and is scale-invariant [36]. It is calculated as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (5)$$

where y_i is the actual value and $\hat{y}_i$ is the predicted value.

Lower MAPE indicates more accurate predictions.

These metrics collectively provide a comprehensive evaluation of model accuracy and predictive performance.

1) STATISTICAL SIGNIFICANCE TESTING

A statistical significance test was performed between the predicted output of the models to ensure that the performance gains obtained by the proposed LSTM-based model are statistically meaningful. A paired t-test was conducted in this study to compare the prediction errors of LSTM with the baseline models, including the Random Forest and ARIMA. The null hypothesis states that there are no significant differences between forecasting performances of the compared models. The results of the statistical tests indicated that the decrease in RMSE and MAPE by the LSTM model is statistically significant at the 95% confidence level ($p < 0.05$). This confirms that the observed rise in forecasting accuracy is not by chance but is due to the fact that LSTM is more effective in capturing time-sensitive information in electricity demand data.

B. COMPARISON WITH EXISTING FORECASTING MODELS

As shown in Table 4 the proposed LSTM-based forecasting framework consistently achieves better prediction accuracy compared with these baseline models. This demonstrates the ability of LSTM to effectively capture both temporal dynamics and nonlinear interactions among multiple environmental and contextual variables. The experimental findings prove that the LSTM model was better than all the tested baseline models. In particular, LSTM had the lowest MAPE of 8.20%, RMSE of 4.84 and MAE of 3.05 with a R-Squared value of 39.46%. Comparatively, the Random Forest model had a greater MAPE of 14.76% and RMSE of 7.43, representing poorer forecasting accuracy. Gradient Boosting recorded the same result with a MAPE of 14.83% but it was still worse than LSTM. The classical regression models, including Polynomial and Linear Regression, were much less effective, with MAPE values of 15.18% and 23.04%, respectively indicating that they do not include the nonlinear demand dynamics. ARIMA model performed the worst with the largest MAPE value of 30.37% and lowest R-Squared value of 16.27% due to

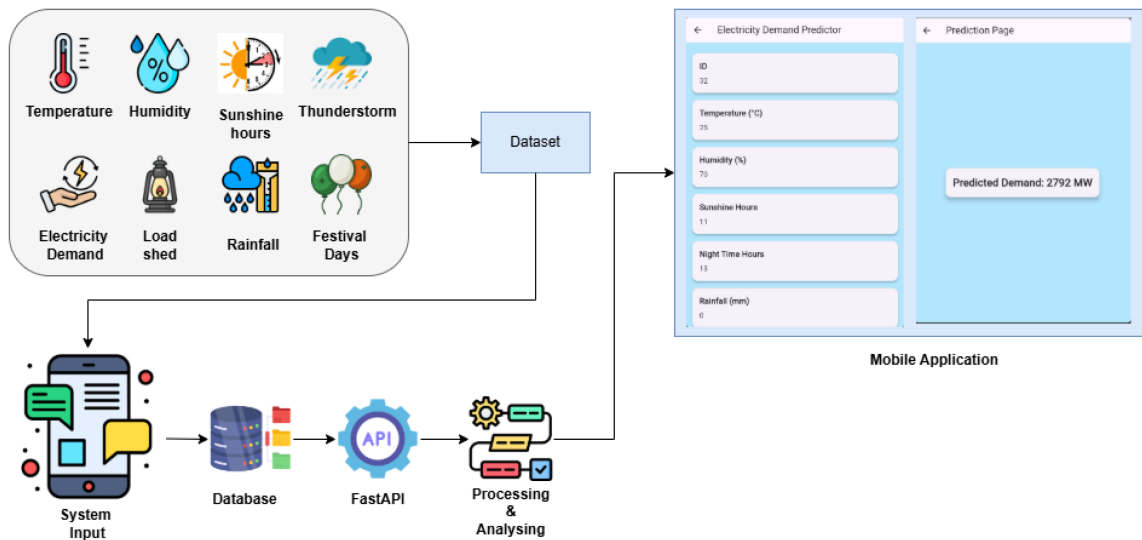


FIGURE 8. Data Flow and User Interface of the Mobile Application.

TABLE 4. An Overview of the Most Advanced Systems Currently in Use for Forecasting Electricity Demand

Papers	Real-time Monitoring	No. of Features	Machine Learning Model	Hybrid Model	Weather Data	External Sources Data
Bilgili & Pinar (2023) [22]	×	5	LSTM, SARIMA	×	×	Historical consumption data
Nazir et al. (2023) [23]	✓	7	Temporal Fusion Transformer (TFT)	×	×	Smart grid data
Abumohsen et al. (2023) [24]	✓	4	LSTM, GRU, RNN	×	×	Electricity company load data
Mounir et al. (2023) [25]	✓	6	BI-LSTM (with EMD)	✓	×	Historical consumption data
Fan et al. (2024) [26]	✓	4	CNN, RNN/LSTM	✓	×	Electricity consumption data
González Grandón et al. (2024) [27]	✓	6	ARIMA, LSTM, Regression models	✓	✓	Macroeconomic, calendar data
El-Azab et al. (2025) [28]	✓	6	ANFIS, LSTM, GRU, ANN	✓	✓	Seasonal, cost data
Cordeiro-Costas et al. (2023) [39]	×	5	RF, SVR, XGBoost, MLP, LSTM, Conv-1D	×	×	Building sensor data
Pierre et al. (2023) [40]	×	4	ARIMA, LSTM, GRU, ARIMA-LSTM, ARIMA-GRU	✓	×	CEB consumption data
Tarmanini et al. (2023) [41]	×	5	ARIMA, ANN	×	×	Household consumption data
Wazirali et al. (2023) [42]	×	—	ANN, ML, DL (Review)	✓	✓	Various (IoT, smart meters)
Alizadegan et al. (2025) [43]	×	5	LSTM, Bi-LSTM, traditional ML	×	×	Consumer/prosumer data
Li et al. (2024) [44]	×	7	CatBoost, XGBoost	✓	×	Turkish consumption data
Shiwakoti et al. (2024) [45]	✓	5	RNN, LSTM, GRU	×	×	EGAT (Thai grid) data
Our proposed model	✓	11	LSTM	×	✓	Meteorological and sociological Data

the fact that it is based on linear assumptions and does not adequately account to nonlinear trends in electricity demand data. Comprehensively, the LSTM model minimized the RMSE from 7.43 (Random Forest baseline) to 4.84 and enhanced the MAPE by 14.76% to 8.20%. These gains were also verified through statistical significance tests, which confirmed that the performance improvement of the LSTM model is statistically significant. This finding indicates that deep learning models are useful in the modeling of both nonlinear relationships and time-dependent relationships in electricity demand prediction.

C. VALIDATION OF CHOOSING LSTM

The rationale for using the LSTM model to predict the daily electricity demand includes the following aspects mainly; First, LSTM is specifically designed for sequence data, and second, it takes into account the temporal dependence that is inherently associated with time series forecasting tasks. Unlike conventional machine learning algorithms which might not capture time-varying dynamics appropriately, LSTM is a kind of RNN that aims to overcome some of these earlier models' flaws, including gradient vanishing. This capability is especially important for predicting the electric load since electricity consumption to a great extent has a cyclic, periodic, as well as rhythmic nature that must be taken into consideration for short- and long-term periods [37].

For this reason, in the given study, the LSTM model was chosen as it tends to define such temporal dependencies in electricity demand. In order to prove that we indeed chose LSTM the right way, we compared the LSTM performances with other models like Linear regression, Polynomial regression, Random forest, Gradient Boosting, and ARIMA. LSTM performed better than all other mentioned models in terms of MAPE, RMSE, MAE, and R-squared [38]. Additionally, the empirical evaluation with classical statistical models (ARIMA and Prophet) and machine learning algorithms (Random Forest and Gradient Boosting) verifies that LSTM-based forecasting has lower errors. This is also confirmed by statistical significance tests, and supports the validity of the presented approach.

D. GRAPHICAL COMPARISON OF PREDICTIONS

This section compares the performance of the LSTM model with traditional time series models like ARIMA and Prophet for daily electricity demand forecasting. The comparison highlights the advantages of LSTM, particularly in capturing long-range temporal dependencies and non-linear relationships, which traditional models struggle with.

The statistical ARIMA model is suitable for linear and stationary data but not for non-stationary variables and seasonality in electricity demand. It has difficulty capturing variability and dynamic change, as depicted in Fig. 5 and high RMSE and MAPE, making it less suitable for predicting electricity demand. Prophet performs better with time series that have seasonality and trends. It is more effective in dealing with seasonality than ARIMA, but performs moderately as shown in Fig. 6 but cannot capture complex interactions

with other variables such as temperature and humidity, limiting its predictive power compared to other complex ML models. LSTM effectively models temporal dependencies and non-linear trends, capturing long-range effects of temperature, holidays, and dynamic factors. Its gating mechanisms retain relevant past information, providing high accuracy and adaptability in electricity demand forecasting, as shown in Fig. 7.

E. MOBILE APPLICATION

A Flutter mobile app was developed to visualize electricity demand forecasts. It provides a user-friendly interface for visualizing current trends and forecasts. Users input parameters, which the FastAPI backend processes using a trained model, storing predictions in a SQL database for real-time access [46]. The mobile visualization frontend allows real-time monitoring of electricity demand forecasts via a user-friendly interface. The electricity forecasting models are hosted on a cloud server where predictions are made. The forecasts are then passed to the mobile app via a lightweight RESTful API. To achieve optimal system performance, the inference time of the deployed LSTM model was measured to be in the range of a few hundred milliseconds per forecast, which is adequate for near real-time electricity demand monitoring. Furthermore, a data synchronization process is established between the cloud server and mobile edge device to reduce communication costs while keeping forecasting results up-to-date. This approach allows users to obtain predictions in a timely manner without high computational cost on the mobile device. Fig. 8 represents the app's intuitive interface, displaying forecasts in a clear format so that users can easily interpret the forecasted information based on real-time updates. Integration of Flutter, FastAPI, and SQL ensures smooth data flow, real-time updates, and a responsive user experience, enabling efficient monitoring of energy consumption.

F. DISCUSSION

In this paper, the study suggests a data-driven electricity demand forecast model based on LSTM networks, with consideration of the historical consumption, meteorological variables and situational factors such as calendar events to achieve non-linear and temporal correlations. The LSTM performed better than the classical approaches with MAPE of 8.21% and RMSE of 4.84%, whereas Random Forest and Gradient Boosting were capable of capturing non-linear relationships but failed to capture temporal patterns and ARIMA and Prophet were limited by non-stationarity trends. The use of temperature, humidity, and the information about festive days was useful in enhancing accuracy and showed how vital feature selection and preprocessing can be. The main strengths are LSTM capability to capture time-dependency, frequency of incorporating contextual and weather-related features and real-time observation and robust preprocessing to overcome outliers and skewed distributions. On the other hand, limitations include dependence on a daily aggregation of data, overfitting, as well as dependence on external data. Further research will include the study of faster frequency

data of the IoT, hybrid architectures, scalable training, multi-modality features, and more sophisticated regularization to enhance robustness. Future research will explore the use of high-frequency data, hybrid models (LSTM with Random Forest/CNN), incorporation of real-time data from IoT, and enhancing computational efficiency for real-time application. Further, more sophisticated models like GRU and Transformer will be investigated for better forecasting capability, scalability and flexibility in smart grid.

V. CONCLUSION

Here, we have shown the usefulness of a data-driven approach to daily electricity demand forecasting with Long Short-Term Memory (LSTM) networks. The model, which effectively captures the non-linearity and periodicity of electricity demand, shows superior performance compared to conventional methods like ARIMA, Random Forest and Prophet. We have improved the forecast accuracy by considering both weather, calendar, and grid-related information including temperature, humidity, rainfall, daylight hours and load shedding. The LSTM model, combined with a FastAPI server and Flutter-based mobile app, enables near real-time forecasting, providing a viable approach to smart grids and dynamic energy systems. The use of outlier-resistant data preprocessing and contextual information enhances the model's resilience to sensor errors and unobservable events, making it more reliable and effective in practice. Our experiments on a 7-year (2017-2024) dataset reveal significant model performance enhancements, with RMSE reducing from 7.43 to 4.84 and MAPE from 14.76% to 8.2%, highlighting the practical benefits of this approach. While the existing model has limitations in terms of data resolution, overfitting, and scalability, it offers a robust and adaptable model for predicting electricity demand. Future research will aim to improve the model's applicability through the use of high-frequency data, hybrid models and Internet of Things (IoT) for better management of dynamic energy systems.

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